

Topic 9 – Forces and their effects

Students should:	Maths skills
9.1 Describe, with examples, how objects can interact - a at a distance without contact, linking these to the gravitational, electrostatic and magnetic fields involved - b by contact, including normal contact force and friction - c producing pairs of forces which can be represented as vectors	
9.2 Explain the difference between vector and scalar quantities using examples	
9.3 Use vector diagrams to illustrate resolution of forces, a net force, and equilibrium situations (scale drawings only)	4a, 5a, 5b
9.4 Draw and use free body force diagrams	4a, 5a, 5b
9.5 Explain examples of the forces acting on an isolated solid object or a system where several forces lead to a resultant force on an object and the special case of balanced forces when the resultant force is zero	5a
9.6P Describe situations where forces can cause rotation	
9.7P Recall and use the equation: moment of a force (newton metre, N m) = force (newton, N) × distance normal to the direction of the force (metre, m)	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
9.8P Recall and use the principle of moments in situations where rotational forces are in equilibrium: the sum of clockwise moments = the sum of anti-clockwise moments for rotational forces in equilibrium	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
9.9P Explain how levers and gears transmit the rotational effects of forces	5b
9.10 Explain ways of reducing unwanted energy transfer through lubrication	

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- **Use vector diagrams to illustrate resolution of forces, a net force, and equilibrium situations (scale drawings only) (4a, 5a, 5b).**

Suggested practicals

- Investigate levers and gears.